

An Analysis of Foundation model AI

Foundation model AI

Foundation model AI -- Part 1

Examples TechCrunch saw Sora as an example of a world model, while in January 2025, Nvidia released its own set of world models. The South China Morning Post wrote that Manycore Tech was another example of companies aiming to build a world model, viewing their work as an example of spatial intelligence. In May 2025, Mohamed bin Zayed University of Artificial Intelligence released a world model for building simulations to test AI agents. Google DeepMind has also released two world models in two-dimensional space and three-dimensional space, respectively, that were trained on video data, with Google claiming that the latter can be a training environment for AI agents. Meta released a world model in June 2025, Tencent released an open source world model in July 2025. Niantic, Inc. spinoff, Niantic Spatial, is developing a world model using anonymized player scans from Pokémon GO. Other companies that are planning as of 2025 to build world models include ByteDance and xAI.

Foundation model AI -- Part 2

Supply chain Foundation models' general capabilities allow them to fulfill a unique role in the AI ecosystem, fueled by many upstream and downstream technologies. Training a foundation model requires several resources (e.g. data, compute, labor, hardware, code), with foundation models often involving immense amounts of data and compute (also referred to as computational power). Due to foundation models' large development costs and inexpensive adaptation requirements, the AI landscape has shifted to a small subset of AI companies making foundation models for downstream adaptation. Thus, most foundation model companies outsource this step to specialized data providers (e.g. Scale AI, Surge) and compute providers (e.g. Amazon Bedrock, Google Cloud, Microsoft Azure).

Foundation model AI -- Part 3

General-purpose AI Due to their adaptability to a wide range of use-cases, foundation models are sometimes considered to be examples of general-purpose AI. In designing the EU AI Act, the European Parliament has stated that a new wave of general-purpose AI technologies shapes the overall AI ecosystem. The fuller structure of the ecosystem, in addition to the properties of specific general-purpose AI systems, influences the design of AI policy and research. General-purpose AI systems also often appear in people's everyday lives through applications and tools like ChatGPT or DALL-E. Government agencies like EU Parliament have identified regulation of general-purpose AI, such as foundation models, to be a high

priority. General-purpose AI systems are often characterized by large size, opacity, and potential for emergence, all of which can create unintended harms. Such systems also heavily influence downstream applications, which further exacerbates the need for regulation. In regards to prominent legislation, a number of stakeholders have pushed for the EU AI Act to include restrictions on general-purpose AI systems, all of which would also apply to foundation models.

Foundation model AI -- Part 4

Applications Fei-Fei Li views world models as applying to robotics and creative works. Due to the complexity of these models, she advocates for more complex strategies in data acquisition, data engineering, data processing, and synthesizing data. She co-founded a startup on building world models, which, as of 2024, planned to do so in three phases: incorporating an understanding of three-dimensional space along with time; support for augmented reality; and support for robotics. Her startup, World Labs, released its commercial world model, Marble, in November 2025. World models are intended for use in interactive media (such as video games and movies) and environment simulation. Proposed use cases for world models include action planning and outcome prediction. Other applications include social simulacra to simulate social systems. Wired compared world models to the metaverse, while Business Insider noted possible military applications. In 2025, world models are being applied to drone warfare, robotics, and self-driving vehicles. The Wall Street Journal speculated that world models could improve spatial reasoning of artificial intelligence models and successfully automate both blue-collar and white-collar jobs. As of October 2025, research has shown mixed results in the spatial reasoning capabilities of text-to-video models (in particular, Veo 3). In Ophthalmology Foundation Models have been applied to retinal imaging including volumetric optical coherence tomography

Foundation model AI -- Part 5

In artificial intelligence, a foundation model (FM), also known as large x model (LxM, where "x" is a variable representing any text, image, sound, etc.), is a machine learning or deep learning model trained on vast datasets so that it can be applied across a wide range of use cases. Generative AI applications like large language models (LLM) are common examples of foundation models. Building foundation models is often highly resource-intensive, with the most advanced models costing hundreds of millions of dollars to cover the expenses of acquiring, curating, and processing massive datasets, as well as the compute power required for training. These costs stem from

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the need for sophisticated infrastructure, extended training times, and advanced hardware, such as GPUs. In contrast, adapting an existing foundation model for a specific task or using it directly is far less costly, as it leverages pre-trained capabilities and typically requires only fine-tuning on smaller, task-specific datasets. Early examples of foundation models are language models like OpenAI's GPT series and Google's BERT. Beyond text, foundation models have been developed across a range of modalities—including DALL-E, Stable diffusion, and Flamingo for images, MusicGen and LLark for music, and RT-2 for robotic control. Foundation models are also being developed for fields like astronomy, radiology, ophthalmology, genomics, coding, time series forecasting, mathematics, and chemistry.

Foundation model AI -- Part 6

Release strategies After a foundation model is built, it can be released in one of many ways. There are many facets to a release: the asset itself, who has access, how access changes over time, and the conditions on use. All these factors contribute to how a foundation model will affect downstream applications. In particular, the two most common forms of foundation model release are through APIs and direct model downloads. When a model is released via an API, users can query the model and receive responses, but cannot directly access the model itself. Comparatively, the model could be directly downloadable for users to access and modify. Both release strategies are often classified as an open release. The exact definition of an open release is disputed, but widely accepted requirements are provided by the Open Source Initiative. Some open foundation models are: PaLM 2, Llama 2, Granite, and Mistral. While open foundation models can further research and development more easily, they are also more susceptible to misuse. Open foundation models can be downloaded by anyone, and particularly powerful models can be fine-tuned to intentionally or unintentionally cause harm. During a closed release, the foundation model cannot be accessed by the public, but is used internally by an organization. Such releases are considered safer, but offer no additional value to the research community or the public at large. Some foundation models like Google DeepMind's Flamingo are fully closed, meaning they are available only to the model developer; others, such as OpenAI's GPT-4, are limited access, available to the public but only as a black box; and still others, such as Meta's Llama 2 are open, with broadly available model weights enabling downstream modification and scrutiny.